

HamidReza Saadi Dadmarzi

Education

- Koç University**, Istanbul, Turkey 2023/11 – now
M.Sc. in Computer Science
Research focus: Password-based authentication systems (MPC), privacy-preserving ML, and post-quantum cryptography.
- Amirkabir University of Technology (Tehran Polytechnic)**, Tehran, Iran 2017/09 – 2020/08
M.Sc. in Applied Mathematics
Research focus: Quantum error correcting codes, Surface codes, information theoretic cryptography.
- Amirkabir University of Technology (Tehran Polytechnic)**, Tehran, Iran 2018/09 – 2020/08
Minor in Physics.
- Amirkabir University of Technology (Tehran Polytechnic)**, Tehran, Iran 2014/09 – 2017/08
B.Sc. in Applied Mathematics
Ranked in the top 5% among all students.

Publications & Research Projects

Publications / Preprints:

- **UpSPA: Secure and Updatable Single Password Authentication** (submitted to ESORICS 2026), ([ePrint version](#), [code](#))
- **RoUpSPA: Robust Updatable Single Password Authentication** (preprint, to be submitted to ACM TOPS)

Selected Research Projects:

- **UpSPA / RoUpSPA: Secure and robust single-password authentication.** Designed and implemented threshold-based password-authentication protocols with registration, authentication, secret-update, and password-update phases. Extended UpSPA with PBB-based synchronization, quorum-based repair, and blockchain anchoring; compared Layer-1 anchoring with Layer-2/rollup-style batching for freshness, rollback resistance, and lower update cost. Also developing the [UpSPA extension](#), an open-source password-manager/browser-extension prototype that avoids a local password vault by distributing protected credential state across storage providers.
- **FLDP / SpectraDP: Frequency-domain differential privacy for federated learning.** Working on spectral masking for differentially private federated learning, comparing frequency-domain selection with random DP baselines under accuracy, privacy, communication, retained-energy, and reconstruction-error metrics.
- **CURE:** Contributed code-level debugging support for a homomorphic-encryption-based split-learning system. Helped investigate Go/Lattigo implementation issues, especially around parallel encrypted experiments; this was implementation support, not paper authorship.
- **QuasarSSO: A post-quantum password-protected SSO system.** Designing a password-protected SSO direction that combines post-quantum cryptographic primitives with practical authentication-system constraints.
- **AEGIS** Studying security-model and implementation issues for recent PQC-IoT handshake protocols, with emphasis on mechanizable analysis and performance-aware design.

Professional Experience

- BTL Coin Crypto Exchange**, Istanbul, Turkey /Remote 2021/05 – 2023/10
Developed personalized market screeners and automated crypto asset management bots using [Golang](#) and [Docker](#).
- Institute for Research in Fundamental Sciences (IPM)**, Tehran, Iran 2020/10 – 2021/03
Optimized quantum kernels for neural networks using [Qiskit](#) and [PyQuil](#).
- Paid University Project**, Tehran, Iran 2019/05 – 2020/07
Developed a data warehouse and integrated data using [Excel](#) and [Tableau](#).

Honors & Awards

- Awarded double major recognition from the Exceptional Talent Center at the university.
- Admitted to the Master of Science program without a national entrance exam, recognized by the Exceptional Talent Center.
- Ranked in the top 5% among all students in the B.Sc. program.

Skills

- **Languages:** Persian (Native), English (Working Professional Proficiency), Turkish (Elementary).
- **Programming:** Rust, Python, Go (Golang), MATLAB.
- **Typesetting:** L^AT_EX.
- **Specific libraries:** Qiskit, PyQuil, Lattigo.
- **Optimization:** `scipy.optimize`, CVXPY.

Academic Services

- External reviewer for: IEEE TIFS (2024–present), ACM ASIACCS (2024, 2026), ES-ORICS (2025, 2026), ACNS (2026), Royal Society Open Science (2024), PeerJ (2026), JISA (2025), and Turkish Journal of Electrical Engineering & Computer Sciences (2025)
- Mentored two interns during Summer 2025; mentoring four interns from Fall 2025 through Summer 2026.

Teaching Experience

Amirkabir University of Technology

- | | |
|---|-----------------------|
| • Time Series Analysis | Fall 2019 |
| • Multivariable Calculus | Spring 2019 |
| • Introduction to C Programming | Fall 2018 |
| • Partial Differential Equations | Fall 2018 |
| • Foundation of Matrix and Linear Algebra | Spring 2018 |
| • One Variable Calculus | Fall 2017 |
| • Introduction to Probability | Spring 2017 |
| • General Physics (Electricity and Magnetism) | Spring 2017 |
| • Foundation of Mathematical Analysis | Spring 2017 |
| • Introduction to Set Theory and Logic | Fall 2016–Spring 2018 |

Koç University

- | | |
|--------------------------------------|--------------------------|
| • Introduction to Python Programming | Fall 2023 |
| • Discrete Structures | Spring 2024 |
| • Data Structures & Algorithms | Fall 2024 |
| • Computer Networks & Security | Spring 2025, Spring 2026 |
| • Modern Cryptography | Fall 2025 |

Conferences & Workshops

- 5th and 6th Frontiers in Mathematical Sciences Conferences, IPM, Tehran, July 2017 and July 2018.
- Workshop on Mathematics of Information-Theoretic Cryptography, [IMS](#), Singapore, September 2016.

References

- Prof. Alptekin Küpçü ([link](#), [email](#))
- Dr. Erfan Salavati ([link](#), [email](#))
- Prof. Masoud Pourmahdian ([link](#), [email](#))